Pole Position for the Neohelikian Nemegosenda Carbonatite Complex, Ontario

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Paleomagnetic data from 17 sites (76 cores; 152 specimens) reveals that the Nemegosenda carbonatite complex retains a primary remanence magnetization of low stability. The pole position of 43° N, 178° W ($P_{0.05}$: 15°, 24°) for the complex is concordant with a group of poles from other rock units in the Superior Province in the age range of 1000 to 1115 m.y. It is concluded that interpreting the 1036 m.y. K-Ar date as the age of emplacement of the complex is in agreement with the mean magnetization direction having been preserved without structural rotation since the body cooled.

Des données paléomagnétiques de 17 sites (76 carottes de forage; 152 specimens) revèlent que le complexe de carbonatite de Nemegosenda conserve un magnétisme rémanent de basse stabilité. La position du pôle à 43° N, 178° W ($P_{0.05}$: 15°, 24°) pour le complexe est concordante avec celles d'un groupe de pôles pour d'autres unités dans la Province Supérieure dont les âges varient entre 1000 et 1115 millions d'années. On peut donc conclure qu'interpréter la date au potassium–argon de 1036 millions d'années comme âge de la mise en place du complexe est un accord avec la direction majeur du magnétisme ayant été preservée sans rotation structurale depuis refroidissement du corps.

[Traduit par le journal]

Introduction

The Nemegosenda carbonatite complex outcrops around Lake Nemegosenda about 20 km northeast of Chapleau, Ontario (Fig. 1). The complex is a nearly circular diatreme or volcanic neck of about 5 km in diameter (Parsons 1961) (Fig. 2). It is known from field relationships to be the last intrusive event into a stable crust of Archean gneisses which outcrop over most of this part of the Superior geologic province. Gittins *et al.* (1967) report minimum K–Ar ages of 988 and 1036 m.y. from slightly altered nepheline from the complex. Neighbouring carbonatite complexes at Seabrook Lake (1103 m.y.), Portage (1069 m.y.), Lackner Lake (1090 m.y.), and Firesand River (1048 m.y.) have similar K–Ar ages using biotite, so that 1036 m.y. appears to be a good estimate for the real age of the Nemegosenda complex. This study was initiated to determine whether the individual phases of the complex could be distinguished by using their paleomagnetic remanence direction, and whether the resultant pole position was concordant with the known upper Neohelikian polar wander curve for the Canadian Shield or whether regional tectonic deformation within the Superior province had caused some pole rotation.

Geology

Parsons (1961) has mapped the geology of the limited outcrop of the Nemegosenda complex (Fig. 2). The Archean gneisses (sites 5, 6, and 17) of the country rock consist of bands of hornblende–oligoclase–andesine and quartz–oligoclase–biotite. The complex has a massive alkaline syenite core composed of variable proportions of nepheline, alkali feldspar, and mafic minerals to form coarse nepheline syenite (site 1), foyaite (sites 2, 8, and 9), juvite (sites 3 and 4), malignite (sites 10 and 14), and pulaskite (site 16). The core is surrounded in part by a contact transition zone of brecciated syenitic rocks with an outward increase in brecciation, alteration, and recrystallization towards the bounding metasomatic aureole. The aureole has an inner pyroxenitic fenite zone (site 15), which reflects dehydration and reduction, and contains magnetiferous garnet and magnetite. The aureole's outer red alkaline fenite zone (sites 12 and 13) reflects hydration and oxidation. The fenite is a dense, brick-red,

TABLE 1. Remanence data by sites

Sample site number	Rock unit	Cores collected	Cores accepted	AF demagnetizing field (oersteds)	Mean Remanence Vector			
					Declination (degrees)	Inclination + down (degrees)	Length R	Precision parameter k
1	Complex core–nepheline syenite	5	4	350	138	77	3.130	3.5
2	Complex core–foyaite	5	5	225	341	65	4.833	24
3	Complex core–juvite	5	4	350	290	17	2.104	1.6
4	Complex core–juvite	5	4	150	165	63	1.306	1.1
5	Archean gneiss	5	2	225	358	56	1.962	26
6	Archean gneiss	5	3	150	324	86	2.925	27
7	Intrusive zone–malignite	3	2	350	278	49	1.919	12
8	Complex core–foyaite	5	4	150	336	49	1.255	1.1
9	Complex core–foyaite	5	4	225	305	33	2.659	2.2
10	Complex core–malignite	5	5	350	322	48	3.382	2.5
11	Intrusive zone–sovite	2	0	225	—	—	—	—
12	Aureole and red alkaline fenite	5	5	350	290	12	4.947	75
13	Aureole and red alkaline fenite	4	3	350	315	−27	2.711	6.9
14	Complex core–malignite	5	3	225	298	12	2.902	20
15	Aureole–pyroxenitic fenite	5	3	150	346	60	3.449	5.5
16	Complex core–pulaskite	5	3	225	302	54	2.494	4.0
17	Archean gneiss	5	4	150	314	64	3.668	9.0



FIG. 1. Location map.

foliated rock that contains hematite. Irregular intrusive, replacement, and rheomorphic lenticular zones of malignite (site 7), jacupirangite, and sovite (site 11) are found in and adjacent to the complex. A few minor late mafic dikes, porphyry dikes, carbonatite dikes, and carbonate veins occur in and adjacent to the complex also. The minor faults have been interpreted from aeromagnetic data.

Procedure

Two to five cores were drilled at each site and then oriented *in situ* using a solar compass. Two specimens were cut from each core giving a total of 158 specimens from 79 cores from 17 sites. After measuring the natural remanent magnetization (n.r.m.) of all specimens using a Schonstedt SSM-1A spinner magnetometer, one pilot specimen per site with typical n.r.m. was step-demagnetized in alternating field (a.f.) peak intensities of 50, 100, 150, 225, 350, 500, 750, and 1000 oersteds (oe) using a Schonstedt GSD-1 AF demagnetizer. Guided by the step-wise changes in remanence direction and intensity of its pilot specimen, an intensity was selected using an objective 'end-point' stability index method (Symons and Stupavsky 1973)[1] in which the remaining specimens from the site were a.f. demagnetized

[1]A rational paleomagnetic stability index (manuscript in preparation).

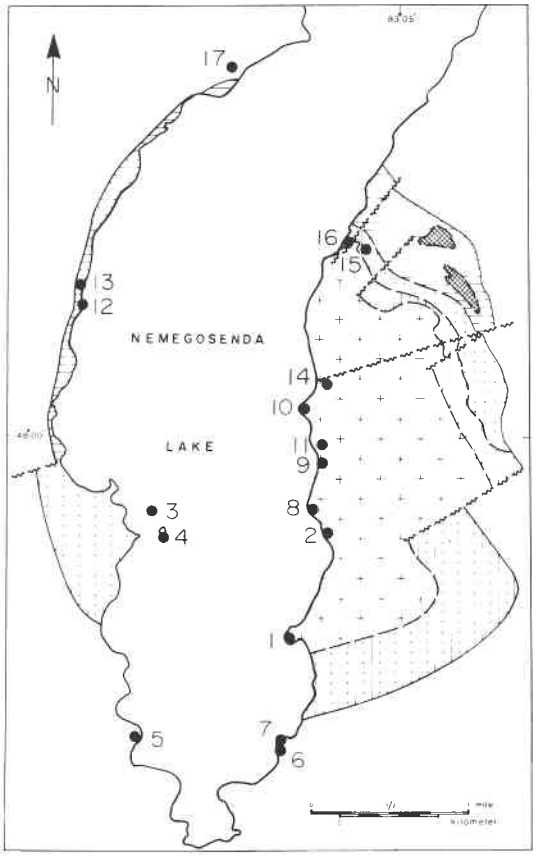


FIG. 2. Nemegosenda carbonatite complex geology after Parsons (1961) and sampling sites. The map units are: alkaline syenite (crosses), syenitic breccia (vertical lines), breccia and pyroxenitic fenite (vertical lines and dots), pyroxenite fenite (dots), red alkaline fenite (horizontal lines), carbonate lenses (crossed hatching), and Archean gneiss (no pattern).

and their residual remanence remeasured (Table 1).

Three factors reveal that a poor to moderately stable characteristic remanence is being isolated in the specimens. First, the median of the mean coercivities determined by a.f. demagnetization for the 14 pilot specimens from the complex sites is a modest 190 oe and for the three Archean gneiss specimens is a low 75 oe. Second, 21 cores or 27% of the collection were rejected as inhomogeneously magnetized because the remanence directions of their two specimens diverged by >20°. After applying this directional criteria only one core remained from site 11 and so it was also re-

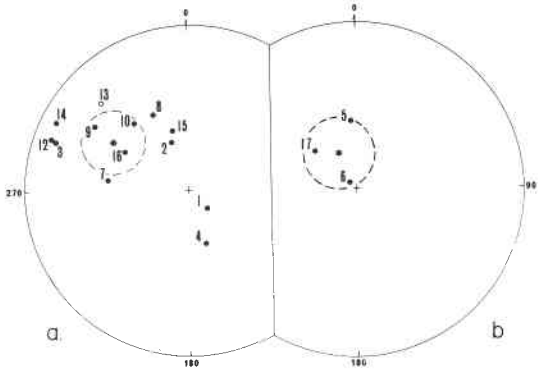


FIG. 3. Schmidt stereographic projection of the site mean remanence directions (Table 1) from the (*a*) Nemegosenda carbonatite complex, and the (*b*) Archean gneiss. The mean remanence direction (star) is surrounded by its cone of 95% confidence ($P_{0.05}$).

jected. Giving each mean core remanence direction unit weight, the mean site remanence directions were then calculated (Table 1). The third factor becomes evident in that the precision parameter values (k of Fisher 1953) for the complex sites are mostly low, thereby indicating a poor to random grouping of directional vectors, and for the Archean gneiss sites they are relatively low thereby indicating a moderate grouping (Fig. 3).

Discussion

Because of remanence instability and resultant scatter in directions, it is clearly not possible to distinguish petrologic phases in this complex using differences in site mean remanence directions.

When all site mean directions from the complex are combined, they give a mean direction of 305°, 46° ($P_{0.05}$: 18°), which gives a pole position of 178° W, 43° N ($P_{0.05}$: 15°, 24°). This position is concordant with pole positions from other rock units of the Superior geologic province dated at between 1000 and 1115 m.y. (Fig. 4). Therefore the paleomagnetic pole position is consistent with the K–Ar age of 1036 m.y. for emplacement of the complex. The pole also indicates that the complex and its surrounding area has not been subjected to more than a few degrees of postemplacement tectonic deformation.

The Archean gneiss gives a mean remanence direction of 338°, 70° ($P_{0.05}$: 18°), which gives a pole position of 141° W, 75° N ($P_{0.05}$:

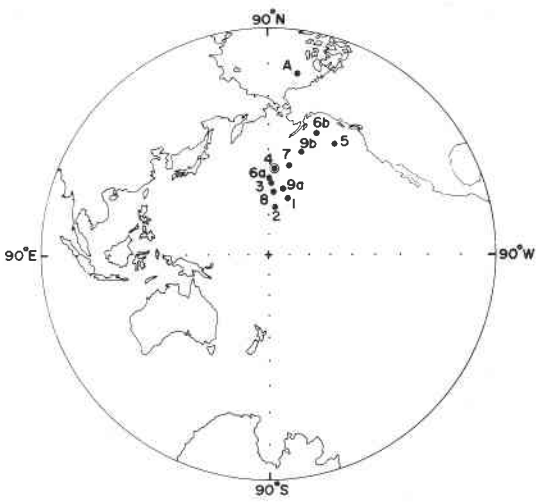


FIG. 4. Pole positions for North American rock formations of the Superior geologic province are plotted on an azimuthal equal-area projection centred on 180° W, 0° N. The poles for the time interval 1000 to 1115 m.y. are: (*1*) Beaver Bay complex, 1000 m.y., (Beck and Lindsley 1969); (*2*) Portage Lake lavas, 1000 m.y., (Vincenz 1968); (*3*) Thunder Bay dikes, 1020 m.y., (Robertson and Fahrig 1971); (*4*) Nemegosenda carbonatite, 1036 m.y., (this study); (*5*) Logan sills, 1050 m.y., (Robertson and Fahrig 1971); (*6*) Maimainse lavas, normal (*a*) and reverse (*b*), 1076 m.y., (Palmer 1970); (*7*) Keweenawan intrusives, 1100 m.y., (Beck 1970); (*8*) Duluth gabbro, 1115 m.y., (Jahrein 1965); and (*9*) North Shore volcanics, normal (*a*) and reverse (*b*), 1115 m.y., (Palmer 1970). *A* is the location of the Archean gneiss pole from this study.

27°, 31°). This position is different from the complex position (Fig. 4), thereby indicating that they have not shared in any postemplacement thermal metamorphic event and strengthening the belief that the complex has retained a primary thermal remanent magnetization of low stability.

Acknowledgments

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Application of Regression Analysis to Interpretation of Geochemical Data from Lake Sediments in Central British Columbia

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Multiple regression is used to investigate relationships among copper, zinc, iron, manganese, organic matter, sand content, and pH of nearshore lake sediment samples from small lakes in central British Columbia. Iron and sand content are shown to be the principal factors influencing regional distribution of copper and zinc.

La régression multiple est utilisée pour fouiller les relations entre le cuivre, le zinc, le fer, le manganèse, la matière organique, le contenu en sable ainsi que le pH des sédiments lacustres près du rivage de petits lacs en Colombie Britannique centrale. Le fer et le contenu en sable se sont avérés être les principaux facteurs influençant la distribution régionale du cuivre et du zinc. [Traduit par le journal]

Introduction

Few geochemical data are available for lakes in western Canada. Consequently, as part of an ongoing program in applied geochemistry, regional variations in trace element content of lake sediments from central British Columbia have been investigated. As part of this study correlation and multiple regression analysis has been used to examine the relationships between pH, organic matter, sand, iron, manganese, copper, and zinc in nearshore sediments.

Description of the Study Area

The study area, covering some 600 mi² (1554 km²) of the Fraser Basin (Holland 1964) northwest of Prince George (Fig. 1), is believed to be underlain by Takla volcanics and the lower Cambrian Snowshoe Formation and Wolverine Complex (Armstrong *et al.* 1968). These units are intruded by Mesozoic quartz diorites and granodiorites. However, due to an almost continuous mantle of glacial till, bedrock is rarely exposed.

Physiographically the region is characterized by low relief with gently rolling hills and a high density of small lakes. Most of these lakes are orientated perpendicular to the northeasterly direction of Pleistocene ice movement (Tipper 1971) and appear to have resulted from ponding by terminal or recessional moraines.

Materials and Methods

During the summer of 1972, Minorex Ltd., Vancouver, collected approximately 1100 lake sediment samples from the region as part of a mineral exploration program. Samples were collected around the margins of lakes using a simple coring device pushed into the lakebed. Where possible, organic-rich material was avoided. From the samples available, 50, randomly distributed over the study area, were selected for detailed investigation.

Samples were air-dried, disaggregated and sieved to −80 mesh for analysis. Copper, iron, manganese, and zinc were determined by atomic absorption after digestion with a mixture of